BatchGuard

Real-time medicine authenticity and batch-risk early warning

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How to use this document. This is a working concept, not a final plan. All scores are my own estimates. Anything tagged **VERIFY** has not been checked against a primary source. Please read it, then answer the questions on the last page - the theme question matters most, because it decides whether we pick this or Zinda.

At a glance

The problem Failed and fake medicines are usually found after the harm, and a copied QR code still looks genuine.	**The idea** Rank what to test first using real CDSCO failure history, watch every verification scan live for clones, and fuse both into one ranked sampling queue for drug inspectors.
Why it can win A real-data core, a strong live demo, a measurable IBM Z result, and a safe decision rule: doubt sends a human to test a sample, it never blocks automatically.	**What is still open** Which theme we are registered under, whether the CDSCO data is usable (go/no-go gate), and whether two layers fit in 24 hours.

1. The problem

- **Coldrif, Sept-Oct 2025.** At least 22 children (some reports say 24) died in Madhya Pradesh and Rajasthan after taking this cough syrup, reported to contain 48.6% diethylene glycol against a 0.1% limit.
- **Slow detection.** Deaths began in September and CDSCO reported the contamination to WHO on 8 October. A central team took 19 samples from the children's doctors and nearby shops; only 4 were declared Not of Standard Quality (NSQ).
- **Nobody can test everything.** India has over 10,000 registered drug manufacturers, and enforcement falls largely on state drug controllers with limited capacity. The real question is what to test first.
- **The data already exists.** CDSCO publishes NSQ and spurious lists every month from central and state labs. Trade press reported 157 NSQ and 2 spurious samples for May 2026, and 194 NSQ and 4 spurious for February 2026.
- **Counterfeits are in the official data.** A spurious drug is made by an unauthorised entity and sold under a legitimate brand; in at least one reported case the real manufacturer confirmed it never made the batch.

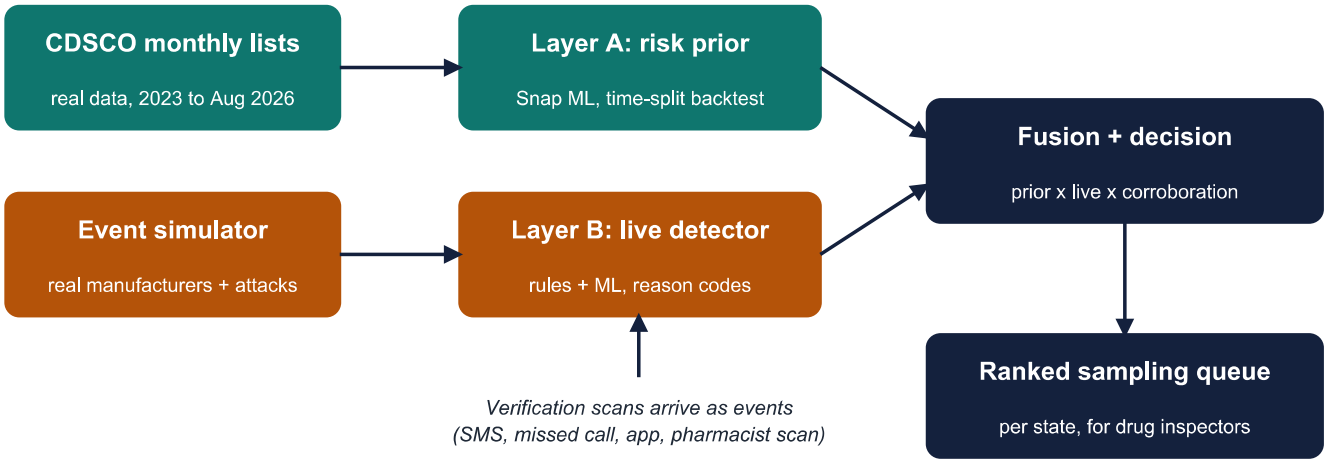
The gap. Lists arrive after the fact and cover only what was tested. A static QR code can be copied onto any number of fake packs. Harmed: patients (especially children), honest manufacturers whose names get attached to fakes, and inspectors with no principled way to prioritise.

2. Why two layers beat one

Layer alone	What it knows	Blind spot	What the other layer adds
Risk prior (history)	Which manufacturers, product forms and states fail most often	Nothing about what is happening right now	A live signal that a specific batch is being cloned or behaving oddly
Live scan detector	How identities behave right now	No idea which batches are inherently risky, so every alert looks equal; can be gamed by scan flooding	Prior risk to rank alerts, and independent evidence to resist flooding

Together they answer a decision a regulator really faces: **which batch should an inspector sample first this week, and why.** The output is an order, not a verdict, because testing capacity is fixed.

3. How it works



Teal = real data. Amber = simulated, clearly labelled. Navy = decision output.

Layer	Job	24-hour deliverable
Data ingestion	Download and parse CDSCO monthly lists (central and state) into one table	Clean table, 2023 to Aug 2026, with source file and month
A. Risk prior	Score manufacturers, product forms and batches for likelihood of failing	Snap ML model, two baselines, time-split backtest
Simulator	Generate scans using real manufacturers, batches and states; inject labelled attacks	Seeded, open generator
B. Live detector	Per-tag state, rule baseline, ML detector, reason codes	Stream service on LinuxONE
Fusion + decision	Combine live anomaly with prior risk; action ladder; ranked queue	Threshold table and queue per state
Privacy + dashboard	Pseudonymise identities, coarsen location; live feed, alerts, queue	Dashboard

4. Layer A: the real-data core

Data. CDSCO monthly lists. The tables I saw carry product, batch number, manufacturing and expiry dates, manufacturer, failure reason, reporting source and lab or state, and month. **VERIFY** the fields and download 12+ months from the actual CDSCO files (my check used news reproductions of the tables).

Features. Manufacturer failure history (counts, recency), product category and dosage form (for example oral liquids, injectables), failure type, state, batch age at test, and a missing-state flag. Use rates wherever a month publishes the number of samples tested.

Model and test. Snap ML gradient boosting against a transparent logistic baseline. Train on lists up to August 2025; test on September 2025 to August 2026. Measure the share of real NSQ and spurious batches that come from the top-k ranked manufacturers or product forms, against (a) random and (b) 'inspect whoever has the most past failures'. Report lift, not accuracy.

Honest limits to state on a slide. Only tested batches appear, so this shows sampling, not the true failure rate - the output is inspection priority, never the probability a batch is unsafe. Some states submit nothing in some months (in August 2024, 18 states and UTs reported no NSQ data - **VERIFY** whether that has improved), so silence is not safety. **Coldrif backtest:** check where the maker would have ranked using data up to August 2025. If it had no earlier failures, say so: history cannot see a first-time failure, which is why the live layer and the product-form prior exist.

5. Layer B: the live scan layer

Event schema. Tag ID, batch ID, timestamp, district, channel (SMS, missed call, app, pharmacist scan), dealer ID, pseudonymous requester ID. **Simulator.** Manufacturers, batches and states come from Layer A, so the stream is anchored in real identities. Attacks are injected at 0.5-2% of events with ground-truth labels. The generator is open and seeded.

Attack	What it looks like	Detection signal
Copied serial	One tag verified in many distant places	Distinct locations, impossible travel
Replay or burst	Same tag verified many times in minutes	Scan rate, inter-arrival time
Guessed serials	Sequential IDs probed by a script	ID entropy, sequence tests, many first-time IDs from one source
Dealer refill and resale	Genuine packs refilled; scans pile up on one dealer after a first valid scan	Post-sale scan pattern per dealer
Slow adaptive clone	Scans spread over time and plausible places to dodge rules	Behaviour model; graph of tags, dealers, districts (stretch)
Scan flooding	Many scans of a real tag from many places to get a genuine batch blocked	Source diversity, rate per requester, corroboration with prior and a physical test

Detectors. A transparent rule baseline, an isolation forest, and a gradient-boosted classifier, each with a plain-language reason code.

Fair evaluation (avoids circular results).

- Hold out whole attack types the model never trained on, and compare rules against ML on those.
- Include the slow adaptive clone as a test the simple rules should fail.
- Report precision, recall, F1, PR-AUC, false alarms per 1,000 genuine scans, latency and events per second. Accuracy is meaningless when clones are rare.
- State plainly that the stream is simulated: it validates the method, not real-world accuracy.

6. Fusion and decision rules

Priority combines three things: the live anomaly score, the batch's prior risk from Layer A, and the number of independent sources behind the alert.

Condition	Action	Who acts
Low anomaly	Accept the verification	Nobody
Moderate anomaly	Warn the buyer: verify with your pharmacist; this batch is under watch	Buyer
High anomaly, or moderate anomaly on a high-prior batch	Add to the ranked sampling queue	State drug inspector
High anomaly plus independent corroboration (unrelated sources or a failed sample)	Escalate with evidence	Regulator

Safety rules. Never block a product automatically on scan data alone - doubt means a human samples and tests. Defend against scan flooding with per-requester rate limits, counts of distinct sources, alert decay, and required corroboration. The queue is budgeted (top N batches per state per week) and every entry carries a reason code, for example: *cloned-scan pattern in 4 districts in 36 hours; maker failed 3 oral-liquid batches in 12 months; no sample tested yet.*

7. Privacy and the IBM Z role

Privacy by design

- Requester IDs and phone numbers pseudonymised with a secret salt
- Location coarsened to district
- Models never see raw personal data
- Stretch: aggregate-only hospital signal for unexplained child kidney-failure clusters, labelled synthetic
- No claim of hardware security modules or Secure Execution unless we actually run them

IBM Z and LinuxONE

- Scans are small, continuous transactions - a natural fit for a high-volume, low-latency platform
- Run the stream and detection on LinuxONE; report events per second and p99 latency against a laptop
- Benchmark Snap ML training and scoring on the same platform
- Snap ML may lack s390x wheels on PyPI - test the conda install at the 16 Oct setup session
- Rule: separate what ran on IBM Z, what ran elsewhere, and what is design intent

8. Why it works

- **The core result is on real data.** The risk layer uses official CDSCO lists with a time-split backtest, while most ideas in this space run on invented data.
- **The layers cover each other's blind spots.** History cannot see a first-time failure or a live clone; scans alone cannot tell which alerts matter.
- **The evaluation is not circular.** Held-out attack types and an adaptive attacker test generalisation, not the generator's own rules.
- **It is safe under doubt.** The decision is a ranked human check, and the design anticipates abuse by scan flooding.
- **It fits the platform and theme.** A live stream with measurable throughput suits IBM Z and matches Real-Time AI for Critical Decisions almost literally.
- **The story is concrete.** Children died while contamination went undetected for weeks, and the counterfeit case is already in the official data.

Novelty caveat: I found no tool that fuses live clone detection with a history-based inspection ranking, but that rests on a few searches and is unconfirmed. Seeds are already covered by the national SATHI QR system, so the focus is medicines, with seeds as a roadmap item.

9. Scorecard (my estimates, 1 to 5)

Criterion	Score	Reasoning
Theme fit	5 / 4	5 under Theme 1; 4 under Theme 2 (privacy layer carries it only partly)
Impact	5	Child deaths, patient safety, honest manufacturers
Data credibility	4	Real CDSCO core; scan stream and hospital signal are synthetic and labelled
IBM Z role	4	Measurable stream and Snap ML results, if they run on LinuxONE
Novelty	4	Not found elsewhere in a few searches; unconfirmed
Technical depth	4	Risk model, stream detector, fusion, adversarial evaluation
24-hour feasibility	3	Two layers is real scope; see the cut list
Defensibility	3	Answers exist, but parsing risk and sampling bias need care

Idea (weighted totals)	Balanced	Judge-style	Risk-first
BatchGuard, if registered under Theme 1	82	83	78
BatchGuard, if Theme 2 (theme score drops to 4)	79	80	76
Zinda with its three fixes (scored for Theme 2)	81	84	75
Federated sepsis early warning (fallback)	80	76	83

The ranking is close, so execution and the theme rules decide it.

10. Risks

Risk	Likelihood	Mitigation
CDSCO files hard to parse, too few usable months	Medium	Run the data gate first; fall back to Zinda
Sampling bias and missing states	Certain	Prioritisation framing, denominators, missing-state flag, say it on a slide
Judges challenge the simulated stream	High	Real identities, open generator, held-out attacks, state the limit
Scope creep across two layers	High	Freeze scope at hour 8; follow the cut list
Adoption depends on QR coverage on packs (mandate not verified)	Medium	Inspector-first pilot; VERIFY what is mandated
Organisers fix the problem statements or theme rules	Medium	Ask before committing; keep Zinda as fallback
Pre-built code not allowed	Unknown	Read the rules; prepare only what is permitted

11. Go / no-go gate (before we commit)

[ ]	Download and parse 12+ months of CDSCO lists; confirm fields and count usable rows.
[ ]	Run the time-split backtest. The ranking must clearly beat 'most past failures', not just random. If not, drop the idea.
[ ]	Confirm Snap ML and a stream service run on the LinuxONE VM (16 Oct, 7:30 PM IST session is the last chance).
[ ]	Confirm with the organisers that our theme and problem statement allow this idea, and that the abstract can change.

If any box fails, we go back to Zinda.

12. The 24 hours

Hours	Work
0-1	Kick-off; check for organiser problem statements or datasets; confirm or adapt
1-5	LinuxONE environment, Snap ML check, load and clean CDSCO data
5-10	Layer A model, baselines, time-split backtest, Coldrif check
10-15	Simulator, stream, rule baseline, ML detector, held-out attack tests
15-19	Fusion, ranked queue, privacy layer, dashboard, throughput and latency runs on IBM Z
19-22	Deck, backup demo recording, README, reproducible notebook
22-24	Rehearse twice, freeze code, submit early and verify

Cut first if behind: graph view, dashboard polish, hospital-cluster module, any second product domain.

13. Demo and pitch

Three-minute demo. (1) Copy a QR from a sample pack and show it still verifies. (2) Replay the stream: clone injected, alert fires with a reason code. (3) Show the ranked sampling queue for one state with prior risk and live evidence. (4) Results slide: lift against baselines, held-out attack performance, false alarms per 1,000 scans, events per second and latency on LinuxONE.

> **Two-minute pitch.** In 2025, at least 22 children died after taking cough syrup with a toxic solvent, and it took weeks to link the deaths to a batch. Regulators publish failed batches every month, but only for what they test, and only afterwards. Meanwhile a copied QR code still calls a fake pack genuine. BatchGuard joins the two: a model on real CDSCO history ranks what to test first, a live engine watches every scan for clones, and when both agree an inspector gets a ranked, explained queue. We never block on doubt - we send a human to test a sample. The risk layer is validated on real data; the scan stream is simulated and we say so. On IBM LinuxONE we measured [X] events per second at [Y] ms p99 latency.

14. Likely judge questions

Question	Suggested answer
Regulators already publish NSQ lists.	After the fact, for tested samples only. We rank before testing and fuse the ranking with live scan evidence.
Is your data real?	The risk layer is real CDSCO data with a time-split backtest. The scan stream is simulated, seeded with real manufacturers, tested on held-out attacks, and labelled.
Isn't NSQ data biased?	Yes: only tested batches appear and some states submit nothing. We use it for prioritisation, show rates with denominators where published, and flag missing states.
Can someone game the alerts?	Scan flooding is a modelled attack. We rate-limit, count independent sources, require corroboration; nothing is blocked automatically.
Why not just QR codes, or SATHI?	A static code is copyable. SATHI covers seeds with QR verification; we focus on medicines and add analytics on top, without claiming SATHI lacks analytics.
What would it have done for Coldrif?	Answer only from the backtest. If the maker had no earlier failures, say history cannot see a first-time failure - hence the live layer and product-form prior.
Why IBM Z?	Measured events per second and latency on LinuxONE, plus Snap ML results; we say what ran there and what we did not test.
Who acts on this?	State drug-controller inspectors through the ranked queue; CDSCO for escalation.
Will anyone actually scan?	Channels are SMS, missed call, app and pharmacist scan. Adoption depends on QR coverage on packs, so we propose an inspector-first pilot.

15. Verify before relying on this

- CDSCO files: fields present, enough months, which states submit, and whether the August 2024 gap has closed.
- Whether QR codes are mandated on the medicines we model (not verified).
- All case statistics against primary sources before they go in a pitch.
- Which Snap ML and LinuxONE capabilities the event environment provides.
- The official judging rubric, submission format, theme rules, pre-event preparation rules, and whether the abstract can change.

Sources reviewed: StarHacks IBM Z Datathon 2026 page; trade-press reproductions of CDSCO monthly NSQ lists (Medical Dialogues, BioPatrika, PharmaTutor); WHO alert and Coldrif reporting (ETV Bharat, Newsreel Asia, Tribune); Deccan Herald on states missing from the August 2024 alert; SATHI portal descriptions (ICRISAT ISSCA, NIC). Check each against the primary source.